

Species-Aware Review: Chromosomal DNA Replication in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu03030` "DNA replication" (18 candidate genes) **Purpose:** Support manual module satisfiability and gene-annotation curation (not a generic pathway essay).

1. Executive summary

The bacterial chromosome-replication module is **fully satisfiable** in *P. putida* KT2440. Every mechanistic step — origin initiation, fork unwinding, single-strand protection, RNA priming, processive Pol III synthesis + proofreading, sliding-clamp processivity, clamp loading, and Okazaki maturation/ligation — is encoded by a confidently identifiable gene. However, the `ppu03030` candidate list is **both incomplete and contaminated**, so it should not be used as ground truth:

- **Incomplete at the initiation step.** The initiator **dnaA (PP_0010)** and the helicase loader **dnaC (PP_1552)** are not in the bucket at all; they live in other KEGG maps. **dnaA** sits in the canonical **dnaA–dnaN–recF–gyrB** origin cluster (PP_0010–PP_0011–PP_0012–PP_0013), which is strong direct genomic evidence.
- **Contaminated by KO over-propagation.** Three genes carry the epsilon KO (K02342) but only **dnaQ (PP_4141)** is the replisome proofreader; **PP_0353** and **PP_4768** are DnaQ/DEDDh exonuclease paralogs. **PP_3893** carries the DnaB KO (K02314) but is a "*putative phage DNA helicase*" — the true replicative helicase is **dnaB (PP_4873)**.
- **Two accessory subunits are unannotatable.** **holD (psi)** and **holE (theta)** have **no ortholog** in KEGG (ppu), the UniProt proteome, or (for psi) a positive-control-validated Pfam/InterPro domain search — and psi is likewise undetectable in *P. aeruginosa* PAO1, i.e. a genus-level pattern. Theta absence is expected (largely enterobacteria-restricted); psi is a

genuine gap/lineage-divergent call. Neither blocks satisfiability (both are non-essential; the core clamp loader DnaX₃-δ-δ'-χ is intact).

- The metadata name "**dnaEA**" for PP_1606 implies a second Pol III alpha. That paralog is **dnaE2/imuC (PP_3119)** within an **imuB–imuC (PP_3118–PP_3119)** SOS mutagenesis cassette — a translesion polymerase, **not** part of the core replisome.

Bottom line for curation: mark all seven module steps **covered**, but re-assign initiation to dnaA/dnaC, restrict the epsilon and helicase steps to the single true gene each, flag PP_0353/PP_4768/PP_3893 as over-propagated paralogs, log psi as a gap and theta as not_expected_in_target_taxon, and move ligB, imuC/dnaE2, rnhA/rnhB, and polA to their more appropriate repair/maturation contexts.

2. Target-organism pathway definition

Included process (KEGG ppu03030, "DNA replication"): the semiconservative synthesis machinery acting at the replication fork — helicase-driven unwinding, single-strand protection, primer synthesis, coordinated leading/lagging-strand Pol III holoenzyme synthesis with proofreading, sliding-clamp processivity, ATP-dependent clamp loading, and RNA-primer removal + nick ligation to mature Okazaki fragments.

Neighboring processes to keep separate: - **Initiation regulation / origin licensing** — dnaA, dnaC, DiaA, Hda/regulatory inactivation. KEGG scatters dnaA/dnaC outside ppu03030; the reusable module should still treat DnaA-dependent origin opening as **step 1**. - **DNA topology** — DNA gyrase (gyrA PP_1767 / gyrB PP_0013), topoisomerase IV (parC PP_4912 / parE PP_4915), topoisomerase I (topA PP_2139, PP_3831). Required for fork progression/decatenation but curated under topology maps, not the replisome step model. - **Mismatch/base-excision/nucleotide-excision repair (KEGG ppu03420/ppu03430)** — polA, ligA, ligB, rnhA, rnhB are placed by KEGG in these buckets even though they physically contribute to lagging-strand maturation. - **SOS/translesion synthesis** — imuA/imuB/dnaE2(imuC), dinB (Pol IV), umuDC/Pol V. These overlap the Pol III alpha family by homology but are a distinct mutagenesis module.

Alternate names / database definitions: KEGG map03030 / ppu03030 "DNA replication"; the replisome is also captured as KEGG modules **M00263** (bacterial DNA polymerase III complex) and by BioCyc/MetaCyc "DNA replication" and Pseudomonas Genome DB (PseudoCAP) categories. UniProt uses the eco/E. coli reference nomenclature (dnaE, dnaQ, holA–holE, dnaX, dnaN).

3. Expected step model (satisfiability call per step)

#	Step	Expected player(s)	KT2440 gene(s)	Call
1	Origin initiation	DnaA (+ DnaC loader)	dnaA PP_0010 (K02313); dnaC PP_1552 (K02315)	covered (outside bucket — metadata gap)
2	Fork unwinding + ssDNA protection	DnaB helicase, SSB	dnaB PP_4873 (K02314); ssb PP_0485 (K03111)	covered (exclude prophage helicase PP_3893)
3	RNA priming	DnaG primase	dnaG PP_0388 (K02316)	covered
4	Pol III core synthesis + proofreading	alpha (DnaE), epsilon (DnaQ)	dnaEA PP_1606 (K02337); dnaQ PP_4141 (K02342)	covered (exclude PP_0353/PP_4768)
5	Sliding-clamp processivity	beta clamp (DnaN)	dnaN PP_0011 (K02338)	covered
6	Clamp loading / replisome coupling	DnaX (tau/gamma), delta (HolA), delta' (HolB), chi (HolC), psi (HolD)	dnaX PP_4269 (K02343); holA PP_4796 (K02340); holB PP_1966 (K02341); holC PP_0979 (K02339); holD = none	covered for core loader; psi = gap/candidate_uncertain
—	(core accessory theta, HolE)	theta	none	not_expected_in_target_taxon
7	Okazaki maturation + nick sealing	Pol I (polA), RNase H (rnhA/rnhB), LigA	polA PP_0123 (K02335); rnhA PP_4142 (K03469); rnhB PP_1605 (K03470); ligA PP_4274 (K01972)	covered

4. Candidate genes and evidence

High-confidence, one-to-one core replisome genes (direct KEGG/UniProt orthology; strong): - **dnaN** / **PP_0011 (P0A120)** — beta sliding clamp (K02338). Origin-cluster context. **Covered, step 5.** - **dnaG** / **PP_0388 (P0A118)** — DNA primase (K02316). **Covered, step 3.** - **ssb** / **PP_0485 (Q88QK5)** — single-strand DNA-binding protein (K03111). **Covered, step 2.** - **dnaEA** / **PP_1606 (Q88MG5)** — Pol III alpha (K02337). The replicative catalytic subunit. **Covered, step 4.** Caveat: "dnaEA" name implies a paralog (see dnaE2 below); confirm this is the *replicative* alpha (it is, by KO K02337). - **dnaQ** / **PP_4141 (Q88FF6)** — Pol III epsilon proofreader (K02342). Adjacent to rnhA (PP_4142). **Covered, step 4.** - **dnaX** / **PP_4269 (Q88F30)** — clamp-loader tau/gamma ATPase (K02343). **Covered, step 6.** - **holA** / **PP_4796 (Q88DM9)** — clamp-loader delta (K02340). **Covered, step 6.** - **holB** / **PP_1966 (Q88LG7)** — clamp-loader delta' (K02341). **Covered, step 6.** - **holC** / **PP_0979 (Q88P74)** — clamp-loader chi (K02339). **Covered, step 6.** - **dnaB** / **PP_4873 (Q88DF2)** — replicative helicase (K02314), GenBank "replicative DNA helicase (ATPase)". **Covered, step 2.**

Okazaki-maturation genes (correct biology, but bucketed elsewhere by KEGG): - **polA** / **PP_0123 (Q88RK6)** — Pol I (K02335) with 5'→3' and 3'→5' exonuclease; removes RNA primers, fills gaps. Primary bucket ppu03420. **Covered, step 7**, cross-listed. - **ligA** / **PP_4274 (Q88F25)** — NAD⁺-dependent DNA ligase (K01972). The **essential** replicative nick-sealer. Primary bucket ppu03420. **Covered, step 7.** - **rnhA** / **PP_4142 (Q88FF5)** — RNase HI (K03469); **rnhB** / **PP_1605 (Q88MG6)** — RNase HII (K03470). Remove RNA primers/R-loops. Supportive of Okazaki processing but also repair; keep as accessory.

Ambiguous / likely over-propagated (see §5).

Evidence type: All assignments above are **homology/orthology-based** (KEGG KO + UniProt), corroborated by **synteny** (origin cluster, dnaQ–rnhA adjacency) and by cross-*Pseudomonas* conservation. No KT2440-specific biochemical/genetic knockouts of individual replisome subunits were found in the literature searched; the strain's genome organization is directly documented (origin/dnaA region conserved with *B. subtilis*,

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5. Gaps, ambiguities, and likely over-annotations

Over-propagated epsilon KO (K02342) — three genes, one true epsilon. Only **dnaQ (PP_4141)** is the replisome proofreader — it is the sole gene carrying the epsilon-specific signatures (CDD **cd06131** "DNA_pol_III_epsilon_Ecoli_like", InterPro **IPR006054 DnaQ + IPR006309 DnaQ_proteo**, PANTHER **PTHR30231:SF41**). **PP_0353** and **PP_4768** are generically named "Exonuclease" and inherit K02342 only because the whole DnaQ/DEDDh 3'→5' exonuclease superfamily maps to Pfam **PF00929 (RNase_T)** / PANTHER family **PTHR30231** / eggNOG **COG0847**. Their discriminating subfamily assignments identify them as *distinct* enzymes: - **PP_0353** → CDD **cd06127** (generic DEDDh), PANTHER **PTHR30231:SF4** "**protein NEN2**" — a NEN/oligoribonuclease-like exonuclease, **not** the replisome epsilon. - **PP_4768** → CDD **cd06127**, PANTHER **PTHR30231:SF37** "**Exodeoxyribonuclease 10**" (**ExoX**) — a mismatch/repair-associated 3'→5' exodeoxyribonuclease. Its UniProt "DNA polymerase III ... core" FUNCTION/SUBUNIT text is transferred boilerplate, not evidence of replisome membership.

→ Both are **candidate_uncertain / remove from module**; do not count toward step 4, and re-annotate with a generic 3'→5' exonuclease term (GO:0008408) rather than the Pol III epsilon term.

Over-propagated DnaB KO (K02314) — prophage helicase. **PP_3893** is a "*putative phage DNA helicase*" sharing K02314 with the real **dnaB (PP_4873)**. Metadata's generic "DNA 5'-3' helicase" label masks its prophage origin. → **not_in_module / candidate_uncertain** (prophage paralog).

Missing accessory subunits. - **hold (psi):** undetectable by **three** independent methods — KEGG KO (K02344), UniProt gene/name search, and a **positive-control-validated domain search** (Pfam **PF03603** / InterPro **IPR004615**, which correctly retrieves *E. coli* hold P28632 but returns nothing in KT2440). The same domain is also absent from *P. aeruginosa* PAO1, so this is a **genus-level pattern**, not a KT2440 annotation lapse. Its chi partner (holC, Pfam PF04364) is present. Interpretation: psi is either genuinely absent in *Pseudomonas* or so divergent that no current family model detects it; because psi is a non-essential accessory clamp-loader subunit and the core loader (DnaX_{3-δ-δ'-χ}) is intact, its absence does **not** block satisfiability. → **gap / lineage-divergent** (resolve only by expert HMM/AlphaFold-structural check). - **hole (theta):** no ortholog; theta is largely enterobacteria-restricted. → **not_expected_in_target_taxon** (expected negative).

Second Pol III alpha = SOS polymerase, not replisome. **dnaE2/imuC (PP_3119, K14162)** with **imuB (PP_3118, K14161)** is the error-prone translesion cassette. The "dnaEA" name in metadata is the tell that a second alpha exists; ensure dnaE2/imuC is curated into the **SOS/ mutagenesis** module, not step 4.

Second ligase. ligB (PP_4968, K01972) shares the LigA KO. LigB is a non-essential NAD⁺-ligase paralog in *Pseudomonas*; **ligA (PP_4274)** is the essential replicative ligase. → keep **ligA** as the step-7 player; mark **ligB** accessory/candidate_uncertain.

Boundary leakage. gyrA/gyrB/parC/parE/topA are required for replication but are topology enzymes; keep them **out** of the replisome step model (separate maps).

6. Module and GO-curation recommendations

Per-step status: - Step 1 initiation → **covered** via dnaA (PP_0010) + dnaC (PP_1552); **module_needs_revision:** add these loci to the reusable module's initiation slot (currently absent from the ppu03030 candidate metadata). - Step 2 unwinding/protection → **covered** (dnaB PP_4873, ssb PP_0485); remove PP_3893 from the module. - Step 3 priming → **covered** (dnaG PP_0388). - Step 4 core synthesis/proofreading → **covered** (dnaEA PP_1606 + dnaQ PP_4141); exclude PP_0353, PP_4768. - Step 5 processivity → **covered** (dnaN PP_0011). - Step 6 clamp loading → **covered** for DnaX/δ/δ'/χ; **psi** = **candidate_uncertain/gap**; **theta** = **not_expected_in_target_taxon**. - Step 7 ligation/maturation → **covered** (ligA PP_4274; polA PP_0123; rnhA/rnhB accessory); ligB accessory.

Module-document actions: - Update bucket boundaries: the reusable module should explicitly place **dnaA/dnaC** (initiation) and, if desired, note topology enzymes as a linked-but-separate module. - Add a **"lineage-specific / expected-absent"** annotation for HolE(theta) so its absence is not scored as an unmet step. - Add a **paralog-disambiguation note** for the epsilon KO (K02342) and DnaB KO (K02314) so future over-propagation is auto-flagged.

GO-curation: - Existing GO terms suffice for all covered steps (e.g., GO:0006269 DNA replication initiation; GO:0006271 DNA strand elongation; GO:0003887 DNA-directed DNA polymerase; GO:0008408 3'-5' exonuclease; GO:0003689 DNA clamp loader; GO:0003896 DNA primase; GO:0003697 ssDNA binding; GO:0003911 DNA ligase (NAD⁺)). **No new GO term requests are needed.** - Recommend GO annotations be **restricted**: e.g., "DNA clamp loader activity" (GO:0003689) should be assigned to the DnaX/HolA/HolB/HolC set, not extended to psi (absent); "3'-5' exonuclease, replisome proofreading" should map to PP_4141 only, with PP_0353/PP_4768 given a generic exonuclease term rather than the Pol III epsilon term.

7. Genes to promote to full fetch-gene review

1. **PP_3893** — resolve prophage vs. functional accessory replicative helicase; likely remove from module. (*High priority.*)
2. **PP_0353** (likely NEN2/oligoribonuclease-like DEDDh exonuclease, PANTHER SF4) and **PP_4768** (likely ExoX / Exodeoxyribonuclease 10, PANTHER SF37) — DnaQ-superfamily exonuclease paralogs mis-KO'd as Pol III epsilon; assign correct specific names, remove Pol III epsilon KO/GO, and re-annotate as generic 3'→5' exonucleases. (*High priority.*)
3. **holD/psi** — three annotation methods (KEGG KO, UniProt name, positive-control Pfam PF03603/IPR004615) all negative in KT2440 and PAO1; only a sensitive profile-HMM/AlphaFold-structural search could distinguish true loss from extreme divergence. (*Medium — expert/structural, not routine fetch-gene.*)
4. **PP_3119 (dnaE2/imuC) + PP_3118 (imuB)** — confirm SOS cassette membership and ensure exclusion from the replicative core step. (*Medium.*)
5. **ligB (PP_4968)** — confirm non-essential paralog status; keep ligA as the module player. (*Low-medium.*)

8. Key references

- Ogasawara N, Yoshikawa H. *Genes and their organization in the replication origin region of the bacterial chromosome*. Mol Microbiol 1992.

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— direct sequence evidence that the *P. putida* dnaA/origin region and gene order are conserved (grounds step 1).

- Takami H et al. *Replication origin region of the chromosome of alkaliphilic Bacillus halodurans C-125*. 1999.

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— notes conserved DnaA-box/origin organization shared with *P. putida* (supports origin-cluster interpretation).

- Fahey J et al. *DinB (Pol IV), ImuBC and RpoS contribute to ciprofloxacin-resistance mutations in Pseudomonas aeruginosa*. 2023.

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- *Pseudomonas* evidence that imuBC/dnaE2 is an SOS mutagenesis (translesion) system, supporting exclusion of PP_3118/PP_3119 from the replicative core (species transfer from *P. aeruginosa* to *P. putida*: strong at the pathway level).
- Molina-Henares MA et al. *Conditionally essential genes for growth of Pseudomonas putida KT2440 on minimal medium*. 2010.

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- KT2440 functional-genomics context (essential-gene framework).
- de Siqueira GMV et al. *Differences in GenBank and RefSeq annotations may affect genomics data interpretation for [P. putida]*. 2025.

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- caution that KT2440 annotation source affects gene calls, directly relevant to over-propagation risk.
- Primary database evidence (direct, this review): **KEGG REST** (rest.kegg.jp) organism `ppu` — KO assignments for all 18 bucket genes, K02313→PP_0010 (dnaA), K02315→PP_1552 (dnaC), K02344/K02345 (holD/holE) → no gene; **UniProt** proteome UP000000556 — no holD/holE entries; gyr/par/top loci.
- Positive-control domain evidence (direct, this review): **UniProt/Pfam/InterPro** — HolD/psi signature **PF03603** / **IPR004615** retrieves *E. coli* K-12 holD (**P28632**) but returns nothing in *P. putida* KT2440 (UP000000556) or *P. aeruginosa* PAO1 (UP000002438); chi signature PF04364 is present (holC=PP_0979). Epsilon discrimination: CDD cd06131 + InterPro IPR006309 (DnaQ_proteo) + PANTHER PTHR30231:SF41 unique to dnaQ=PP_4141; PP_0353=SF4 (NEN2-like), PP_4768=SF37 (ExoX).

Uncertainty & species-transfer notes

- All KT2440 gene→step assignments are **homology/orthology-based** (KEGG KO + UniProt) plus **synteny**; no strain-specific subunit knockouts were located. Confidence is high for the universally conserved core (alpha, beta, DnaB, SSB, DnaG, DnaX, LigA) and high that PP_0353/PP_4768/PP_3893 are paralogs (based on KO sharing + generic/prophage names).
- HolE(theta) absence is a well-established phylogenetic pattern (enterobacteria-restricted); transfer to KT2440 as an expected negative is **strong**.

- HolD(psi) absence is now supported by **three** independent methods (KEGG KO, UniProt name, and a positive-control-validated Pfam/InterPro domain search) and is a **genus-level** pattern (also undetectable in *P. aeruginosa* PAO1 while robustly found in *E. coli*). This strongly indicates either true loss or extreme divergence; final discrimination requires profile-HMM/AlphaFold-structural analysis. It does not affect satisfiability (psi is a non-essential accessory subunit; core loader intact).
- imuBC/dnaE2 role is transferred from *P. aeruginosa*/general *Pseudomonas* SOS biology — **strong** at the pathway level, not yet demonstrated biochemically in KT2440.

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